

Zhaoxin Feng

Department of Language Science and Technology
The Hong Kong Polytechnic University
Hong Kong SAR, China

zhaoxinbetty.feng@connect.polyu.hk
<https://zhaoxin-feng.github.io/>

Education

[†] Indicates expected

- 2024–2028 [†] Ph.D., The Hong Kong Polytechnic University
Supervisor: Emmanuele Chersoni, Co-supervisor: Huang Chu-Ren
- 2027–2028 [†] Visiting Ph.D. Student, University of California, San Diego
Supervisor: Julian McAuley
- 2023 Spring Exchange Student, The Hong Kong Polytechnic University
- 2020–2024 B.Sc., Educational Technology (Minor: Data Science), Beijing Normal University
GPA: 3.7/4.0 (top 7%); Outstanding Undergraduate Award

Research Overview

My research asks a central question: can we rely on LLMs and LLM-based agents when they interact with humans? I study this at two levels: how models represent meaning internally, and how they behave when interacting with people.

- **Internal representations (reliability and interpretability).** I probe how LLMs represent human language: whether enabling bidirectional attention in autoregressive LLMs reshapes their lexical semantic representations, and how their embeddings compare to human mental lexicon judgments. The question is whether a model’s internal grasp of meaning is faithful enough to be relied on.
- **Behavioral reliability in interaction.** I study how LLMs and agents behave with human users, from sycophancy that chain-of-thought reasoning both mitigates and masks, to reasoning failures on Chinese phonological ambiguities, to memory agents that recall user preferences but fail to act on them.

In the long run, I hope to contribute to **personalization**: a personal, long-term agent that deepens its memory through continuous interaction with a user, one that truly understands your context and stays reliable as it does so. My PhD thesis, *Rethinking Memory for Long-Term LLM Agents: Not Just Remembering, But Acting*, takes a step toward this through agentic memory: how preferences, explicit and implicit, are extracted from everyday interactions; how memory is retrieved, updated, and forgotten; and how to evaluate not just what an agent stores but whether it reliably acts on it.

Publications

* Indicates co-first author

Submissions Under Review

- [1] Ma, J., **Feng, Z.**, Chersoni, E., & Chen, S. Every time I hire a linguist, inference costs go down: On linguistic rules as effective prompt compressors. Submitted to *ACL Rolling Review* (Submitted to EMNLP2026).
- [2] **Feng, Z.**, Ma, J., & Chersoni, E. Know it, act on it: Investigating memory utilization in LLM personalization. Submitted to *ACL Rolling Review* (Submitted to EMNLP2026).

Conference Papers

- [1] Chen, Z., **Feng, Z.**, Yip, T. P., Ma, J., Chersoni, E., & Li, B. (2026). PCA-guided activation scaling for monotonic bidirectional control over LLM sycophancy. In *Proceedings of the Conference on Language Modeling (COLM 2026)*.
- [2] **Feng, Z.**, Chen, Z., Ma, J., Yip, T. P., Chersoni, E., & Li, B. (2026). Good arguments against the people pleasers: How reasoning mitigates (yet masks) LLM sycophancy. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (ACL 2026)*.
- [3] Ma, J.^{*}, **Feng, Z.**^{*}, Chersoni, E., Song, H., & Zhang, Z. (2025). PhonoThink: Improving large language models' reasoning on Chinese phonological ambiguities. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP 2025, Oral)*.
- [4] **Feng, Z.**, Ma, J., Chersoni, E., Zhao, X., & Bao, X. (2025). Learning to look at the other side: A semantic probing study of word embeddings in LLMs with enabled bidirectional attention. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL 2025, Oral)*.
- [5] Song, H., **Feng, Z.**, Chersoni, E., & Huang, C.-R. (2025). Which model mimics human mental lexicon better? A comparative study of word embedding and generative models. In *Proceedings of the 16th International Conference on Computational Semantics (IWCS 2025, Oral)*.
- [6] Ma, J.^{*}, **Feng, Z.**^{*}, Song, H., Chersoni, E., & Chen, Z. (2025). Reasoning or memorization? Investigating LLMs' capability in restoring Chinese internet homophones. In *Proceedings of the 3rd Workshop on Towards Knowledgeable Language Models (KnowFM 2025)*.
- [7] Chen, Z., **Feng, Z.**, Ma, J., Xu, J., & Li, B. (2025). Can LLMs recognize their own analogical hallucinations? In *Proceedings of the 3rd Workshop on Towards Knowledgeable Language Models (KnowFM 2025)*.

Selected Coursework

- 2024–Present **Ph.D.-level:** Computational Linguistics; Advanced Natural Language Processing; Advanced Topics in Computer Algorithms; Advanced Large-Scale Machine Learning Systems for Foundation Models; Artificial Intelligence
- 2020–2024 **Bachelor-level:** Single Variable Calculus; Multivariable Calculus and Linear Algebra; Discrete Mathematics; Probability Theory and Mathematical Statistics; Stochastic Process; Deep Learning; Mathematical Modeling; Computer Network Programming; Data Structure; Introduction to Pattern Recognition; Software Engineering; Educational Application of AI

Selected Honours and Awards

- 2026 PolyU PhD Scholars International Collaborative Research Fellowship (ICRF)
- 2025 HKSAR Talent Development Scholarship
- 2024 Outstanding Undergraduate Graduate Award, Beijing Normal University
- 2024 Honourable Mention, Interdisciplinary Contest in Modeling (ICM)
- 2023 PolyU Presidential PhD Fellowship
- 2022 First-Class Academic Scholarship, Beijing Normal University
- 2022 First-Class Competition Scholarship, Beijing Normal University
- 2022 Merit Award for Outstanding Students, Beijing Normal University
- 2022 Second Prize, 15th National Undergraduate Mathematics Competition
- 2022 Gold Award, International Genetically Engineered Machine Competition (iGEM)
- 2022 Volunteer Service Award

Teaching

- 2026 Spring Teaching Assistant, Programming and Data Analysis for Language Studies
- 2025 Fall Teaching Assistant, Computer Programming in Language and Communication
- 2025 Spring Teaching Assistant, Programming and Data Analysis for Language Studies

Research and Industry Experience

- [1] **ByteDance**, Beijing, China *Feb 2024 – Jun 2024*
Strategy Product Manager Intern (AI Application), ZERO Team. Developed product strategy for *Doubao* across OCR, retrieval-augmented generation (RAG), plugins, and agentic workflows in the scenario of education.

- [2] **Polytechnique Montreal**, Montreal, Canada *May 2023 – Aug 2023*
 Research Intern. Led the project named *Understanding the Characteristics of Internet Slangs in OSI Discussions*; and attended another project related to Human Computer Interaction, responsible for integrated physical objects into AR interaction and developing virtual-button functionality in Unity.
- [3] **Institute of Psychology, Chinese Academy of Sciences**, Beijing, China *Oct 2022 – Jan 2023*
 Research Intern. Conducted data analysis and text mining on a dataset of over 50,000 rows using Python and R.

Where I Was and Will Be

2026	Deeplearn 2026, Orléans, France
2026	ACL 2026, San Diego, US
2025	IWCS 2025, Düsseldorf, Germany
2025	EMNLP 2025, Suzhou, China
2025	ACL 2025, Vienna, Austria